Helm Interview Questions & Answers

1. What is Helm in Kubernetes?

Helm is a package manager for Kubernetes that helps define, install, and manage applications using reusable YAML templates called Charts.

2. What are Helm Charts?

A chart is a bundle of YAML files that describe a Kubernetes application, including templates, configurations, and metadata.

3. What is a Helm Release?

A release is an instance of a chart deployed to a Kubernetes cluster. Multiple releases of the same chart can exist with different configurations.

4. What’s the difference between 'helm install' and 'helm upgrade'?

'helm install' is used to deploy a new release, while 'helm upgrade' updates an existing release with new configurations.

5. How do you roll back a Helm deployment?

Use 'helm rollback ' to revert your release to a previous version.

6. What is values.yaml used for?

This file stores configuration values that control how templates are rendered. You can override them using '--set' or another YAML file.

7. What are Helm Repositories?

A Helm repository is a collection of charts stored remotely or locally, similar to how Docker images are stored in DockerHub.

8. How do you list all Helm releases in a namespace?

Use 'helm list -n '.

9. What’s the use of 'helm lint'?

It checks charts for syntax or logical errors before deployment. It’s a good validation step before production deployment.

10. Why use Helm?

Helm simplifies deployments, enables version control, promotes reusability with charts, and integrates well with CI/CD pipelines.

11. What are Helm Hooks?

Helm hooks allow you to run certain Kubernetes resources at specific lifecycle events, like pre-install or post-upgrade.

12. What’s the purpose of 'helm dependency update'?

It downloads and updates chart dependencies listed in the Chart.yaml file.

13. What’s the difference between Helm 2 and Helm 3?

Helm 3 removed Tiller, improved security, added better chart management, and integrated deeply with Kubernetes RBAC.

14. How do you delete a Helm release?

Use 'helm uninstall ' to delete a release from your cluster.

15. How can you view the revision history of a Helm release?

Use 'helm history ' to see all revisions and details of a release.

16. How does Helm handle versioning?

Each chart has a version defined in Chart.yaml, and each deployment (release) has its own revision number tracked by Helm.

17. What are subcharts in Helm?

Subcharts are dependent charts packaged inside a parent chart to modularize large applications.

18. How can you test a Helm chart before deploying?

Use 'helm template' to render manifests locally or 'helm lint' to validate the chart’s structure.

19. What are common Helm commands every DevOps engineer should know?

'helm install', 'helm upgrade', 'helm list', 'helm uninstall', 'helm rollback', 'helm lint', 'helm repo add'.

20. How does Helm support CI/CD pipelines?

Helm charts can be automated in CI/CD pipelines to manage deployments, rollbacks, and versioning seamlessly.